

Major Project

28/09 - 04/10

(W1)

- Carried out beginnings of literature review, read:
 - Mehta & Bradshaw (1979), "Design rules for small low speed wind tunnels". pages 443-449
 - Bell & Mehta (1988), "Contraction design for small low-speed wind tunnels", NASA report, 37 pages.
 - Cooper & Mason (1982), "Correlation Experience with the SAE Wind Tunnel Test Procedure for Trucks and Buses"
 - ~~Griffin~~ Gattin, (1988), "Wind Tunnel Investigation of the Low Speed Aerodynamics of Slender Accelerator-Type Configurations".
 - NASA Glenn's Beginner Guide to Wind Tunnels
- Through research identified couple things:
 - aircraft tunnels tend to be big, I assumed this was due to angle of attack however it is actually due to the Reynolds number being close to real flight.
 - ↳ everything in the tunnel depends on Reynolds number
$$= (\text{air density} \times \text{speed} \times \text{length}) / \text{viscosity}$$
 - however, when looking at cars there is another problem
 - ↳ Ground simulation - under a real ~~road~~ car the road is still relative to air, in a tunnel the ~~car~~ floor builds up it's own "boundary layer" of air under the model.
 - ↳ car tunnels use belts or suction to fix this, which adds cost.
- To fix, maybe add subject to platform with perforations to allow air flow?
 - ↳ ridged edge to prevent air flowing over platform.

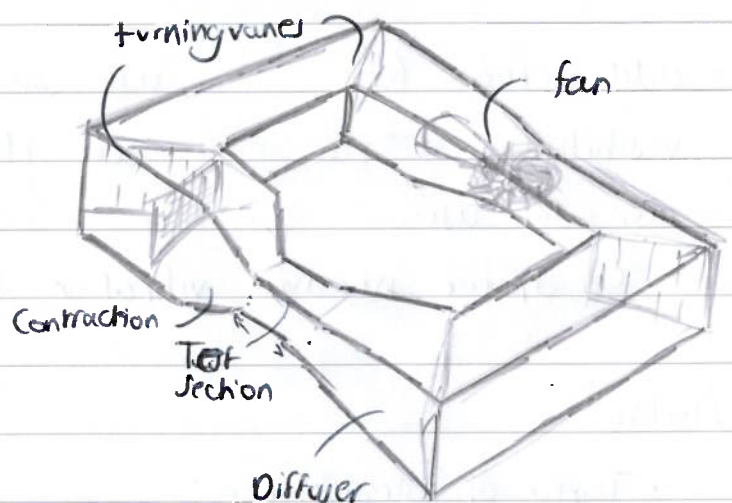
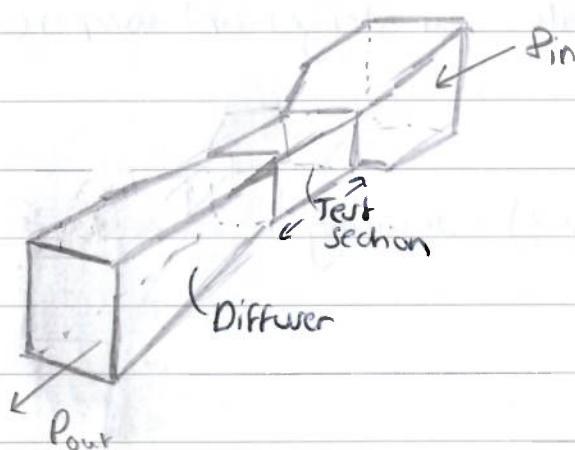


- There are many different types of wind tunnel, categorised by speed regime, circuit design and orientation (special purpose).

↳ speed regime (Mach number):

- Subsonic : $M < 0.8$, used for low speed testing of cars and general aircraft.
- Transonic : $0.8 < M < 1.2$, near the speed of sound where shockwaves form
- Supersonic : $1.2 < M < 5.0$, used for high speed fighter jets and missiles.
- Hypersonic : $5.0 < M$, used for space re-entry vehicles and extreme thermal testing.

- circuit configuration:



Open-circuit tunnel

- draws room air from environment through open end and exhausts back out.
- can have fan at inlet or outlet

Closed-circuit tunnel

- recirculates same air continuously through a looped tunnel circuit, offering better energy efficiency and flow control
- turning vanes require immense accuracy (difficult to get right)

- tunnels can be horizontal or vertical.

- Meeting with supervisor (20/09/26):
 - need a fan, depends on cross section (research other properties?)
 - straightener needed at inlet to reduce turbulence.
 - research benefits of circular tunnels (if any)
 - lengths, types of straightener etc.
 - sensor for drag + lift (within 10% accuracy is successful error)
 - ↳ research inaccuracies.
 - ↳ sensor should be stable
 - consider design parameters
 - ↳ budget is £200/250
 - safety and risk assessment needs doing in box format
- add screen for visibility
 - use small scale car designs and compare air flow
- consider use
 - ↳ space needed, vibration (clamp?) - sewing.

(01/10/26)

• Types of straightener:

Type		Advantages	Disadvantages
Hexagonal honeycomb		Lowest pressure loss of the common shapes. Strong + stiff, easy to buy as aluminium or polycarbonate sheet.	Bought sheet costs money. Cutting it to fit a round or odd-shaped section is fiddly.
Square cells		Easy to make, either by 3D printing or from slotted sheets. Simple to fit to a square settling chamber.	Pressure loss is a bit higher than hexagonal, corners can trap slower flow.

Type	Advantages	Disadvantages
Circular tubes (e.g. drinking straws)	Very cheap + good for a small budget, easy to pack to any cross section	Highest loss of the 3. The gaps between tubes are uneven and add to the loss, hard to keep them aligned.
3D printed (any shape)	Any geometry, fits the section exactly, and is free to make (in-house)	Minimum wall thickness lowers porosity, so loss goes up. FDM surface roughness and layer lines can add turbulence.
Perforated plate/pipe flow conditioners	Made for pipe flow and compact	Mostly for flow metering in pipes. High loss, so rarely used in wind tunnels

- honeycomb mainly removes swirl and sideways velocity - does much less to ~~side~~ fluctuations along the flow, which is why fine screens are usually placed downstream.
- common rule of thumb is L/D of around 6-8, around 150 cells across the section, and porosity of at least 0.8 (Mehta & Brackshaw)
- circular tunnel vs square/rectangular:

↳ Circular

• Advantages:

- No corners, so no corner vortices, boundary layer grows evenly around the wall.
- Construction is symmetric, simpler to design and analyse.
- Structurally efficient, wall stays rigid under pressure difference

• Disadvantages:

- No flat floor, matters if floor suction or ground plane is needed.
- Curved windows distort the view.
- Models and probe traverses are harder to mount + seal,

→ harder to make by hand, printing in segments + adding joints
→ Square/rectangular:

• Advantages:

- Walls are flat so can add cheap sheet material and windows easily.
- Flat floor for section
- Mounting traversing + access are easier

• Disadvantages:

- Corners produce secondary flows, boundary layers thicken here
- Corners of a square are prone to local separation.

W2

05/10 - 11/10

• new website built (05/10/26) - josephbradshaw.github.io

↳ coded on Github, public access, contains:

- CV
- project title and evidence
- logbook entries
- reports

• allows logbook backup, can now ~~also~~ manage risk of losing logbook.

• meeting on 06/10:

- discussed research + progress on design.

↳ types of straightener, circular vs rectangular tunnels.

- Second project idea suggested:

↳ current project ^{could be} difficult, so alternative suggested. University has wind tunnel, could design attachment that allows testing of regular samples.

↳ no need to "create" new wind tunnel, just attachment + sensors etc.

- tunnel is 2m long